THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 4	$a_2 - a_1 = 4$
2	**(a)** 39	$a + (n-1)d = 39$
3	**(b)** 55	$S_n = \frac{n}{2}[2a + (n-1)d]$
4	**(c)** $4, 7, 10, 13, \ldots$	Only $4, 7, 10, 13, \ldots$ has the same difference 3 between consecutive terms
5	**(c)** 8	$a_2 = a + d$
6	**(d)** 3	$a_3 = a + 2d$
7	**(b)** -3	$a_2 - a_1 = -3$
8	**(a)** 26	$a + (n-1)d = 26$
9	**(b)** 72	$S_n = \frac{n}{2}[2a + (n-1)d]$
10	**(d)** $9, 7, 5, 3, \ldots$	Only $9, 7, 5, 3, \ldots$ has the same difference -2 between consecutive terms

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$5, 8, 11, 14, \ldots$ (2).	2
12	$d = a_2 - a_1 = 11 - (6)$ (1) $= 5$ (1).	1+1
13	$d = 9$ constant (1); next term 38 (1).	1+1
14	$10, 8, 6, 4, \ldots$ (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a = 3$, $d = 4$ (1); $a_n = a + (n-1)d$ (1); $a_{12} = 3 + 11(4) = 47$ (1).	1+1+1
16	$a = 2$, $d = 3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{10} = \frac{10}{2}[4 + 9(3)] = 155$ (1).	1+1+1
17	$a_n = a + (n-1)d$: $54 = 4 + (n-1)(5)$ (1); $n - 1 = 10$ (1); $n = 11$, so it is the 11th term (1).	1+1+1
	OR: $a = 5$, $d = 3$ (1); $a_n = a + (n-1)d$ (1); $a_{10} = 5 + 9(3) = 32$ (1).	1+1+1
18	$a = 5$, $d = 5$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_8 = \frac{8}{2}[10 + 7(5)] = 180$ (1).	1+1+1
	OR: $a_n = a + (n-1)d$: $59 = 3 + (n-1)(7)$ (1); $n - 1 = 8$ (1); $n = 9$, so it is the 9th term (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$a = 7$, $d = 3$, $l = 64$ (1); $l = a + (n-1)d$ gives $n - 1 = 19$ (1); $n = 20$ (1).	1+1+1
	OR: $a_n = 6 + 19(4) = 82$ (1); $S_n = \frac{20}{2}(6 + 82) = 880$ (2).	1+2
20	$a = 5$, $d = 5$, $l = 80$ (1); $l = a + (n-1)d$ gives $n - 1 = 15$ (1); $n = 16$ (1).	1+1+1
	OR: $a_n = 9 + 13(3) = 48$ (1); $S_n = \frac{14}{2}(9 + 48) = 399$ (2).	1+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 6	$a_2 = a + d$
2	**(b)** 5	$a_3 = a + 2d$
3	**(a)** 5	$a_2 - a_1 = 5$
4	**(c)** 37	$a + (n-1)d = 37$
5	**(b)** 105	$S_n = \frac{n}{2}[2a + (n-1)d]$
6	**(a)** $1, 6, 11, 16, \ldots$	Only $1, 6, 11, 16, \ldots$ has the same difference 5 between consecutive terms
7	**(c)** 15	$a_2 = a + d$
8	**(c)** 6	$a_3 = a + 2d$
9	**(a)** 6	$a_2 - a_1 = 6$
10	**(c)** 30	$a + (n-1)d = 30$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d = a_2 - a_1 = 17 - (20)$ (1) $= -3$ (1).	1+1
12	$d = -4$ constant (1); next term -1 (1).	1+1
13	$7, 11, 15, 19, \ldots$ (2).	2
14	$d = a_2 - a_1 = \frac{11}{2} - (4)$ (1) $= \frac{3}{2}$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a = 2$, $d = 7$ (1); $a_n = a + (n-1)d$ (1); $a_8 = 2 + 7(7) = 51$ (1).	1+1+1
16	$a = 1$, $d = 4$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{12} = \frac{12}{2}[2 + 11(4)] = 276$ (1).	1+1+1
17	$a_n = a + (n-1)d$: $54 = 2 + (n-1)(4)$ (1); $n - 1 = 13$ (1); $n = 14$, so it is the 14th term (1).	1+1+1
	OR: $a = 10$, $d = 5$ (1); $a_n = a + (n-1)d$ (1); $a_{15} = 10 + 14(5) = 80$ (1).	1+1+1
18	$a = 7$, $d = 2$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_{15} = \frac{15}{2}[14 + 14(2)] = 315$ (1).	1+1+1
	OR: $a_n = a + (n-1)d$: $31 = 10 + (n-1)(3)$ (1); $n - 1 = 7$ (1); $n = 8$, so it is the 8th term (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$a = 2$, $d = 6$, $l = 68$ (1); $l = a + (n-1)d$ gives $n - 1 = 11$ (1); $n = 12$ (1).	1+1+1
	OR: $a_n = 2 + 17(5) = 87$ (1); $S_n = \frac{18}{2}(2 + 87) = 801$ (2).	1+2
20	$a = 10$, $d = 4$, $l = 106$ (1); $l = a + (n-1)d$ gives $n - 1 = 24$ (1); $n = 25$ (1).	1+1+1
	OR: $a_n = 15 + 24(2) = 63$ (1); $S_n = \frac{25}{2}(15 + 63) = 975$ (2).	1+2

Mistakes to avoid:

- Using $a_n = a + nd$ instead of $a + (n-1)d$.
- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
- Using S_n when only a_n is asked (or the other way round) in word problems.
- Not rejecting a negative or fractional value of n obtained from a quadratic in n.
- Calling a list an AP after checking only the first two differences.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 5: **Arithmetic Progressions** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 25	$S_n = \frac{n}{2}[2a + (n-1)d]$
2	**(a)** $6, 12, 18, 24, \ldots$	Only $6, 12, 18, 24, \ldots$ has the same difference 6 between consecutive terms
3	**(c)** 12	$a_2 = a + d$
4	**(d)** 2	$a_3 = a + 2d$
5	**(a)** $\frac{1}{2}$	$a_2 - a_1 = \frac{1}{2}$
6	**(d)** 37	$a + (n-1)d = 37$
7	**(a)** 30	$S_n = \frac{n}{2}[2a + (n-1)d]$
8	**(d)** $-3, 1, 5, 9, \ldots$	Only $-3, 1, 5, 9, \ldots$ has the same difference 4 between consecutive terms
9	**(a)** 11	$a_2 = a + d$
10	**(b)** -4	$a_2 - a_1 = -4$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$d = 6$ constant (1); next term 27 (1).	1+1
12	$-3, 2, 7, 12, \ldots$ (2).	2
13	$d = a_2 - a_1 = -4 - (-8)$ (1) $= 4$ (1).	1+1
14	$d = 7$ constant (1); next term 39 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$a = 1$, $d = 6$ (1); $a_n = a + (n-1)d$ (1); $a_9 = 1 + 8(6) = 49$ (1).	1+1+1
16	$a = 3$, $d = 6$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_9 = \frac{9}{2}[6 + 8(6)] = 243$ (1).	1+1+1
17	$a_n = a + (n-1)d$: $77 = 5 + (n-1)(6)$ (1); $n - 1 = 12$ (1); $n = 13$, so it is the 13th term (1).	1+1+1
	OR: $a = 4$, $d = 2$ (1); $a_n = a + (n-1)d$ (1); $a_{20} = 4 + 19(2) = 42$ (1).	1+1+1
18	$a = 10$, $d = 3$ (1); $S_n = \frac{n}{2}[2a + (n-1)d]$ (1); $S_6 = \frac{6}{2}[20 + 5(3)] = 105$ (1).	1+1+1
	OR: $a_n = a + (n-1)d$: $49 = 1 + (n-1)(8)$ (1); $n - 1 = 6$ (1); $n = 7$, so it is the 7th term (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$a = 4$, $d = 7$, $l = 102$ (1); $l = a + (n-1)d$ gives $n - 1 = 14$ (1); $n = 15$ (1).	1+1+1
	OR: $a_n = 8 + 11(6) = 74$ (1); $S_n = \frac{12}{2}(8 + 74) = 492$ (2).	1+2
20	$a = 3$, $d = 9$, $l = 84$ (1); $l = a + (n-1)d$ gives $n - 1 = 9$ (1); $n = 10$ (1).	1+1+1
	OR: $a_n = 4 + 29(3) = 91$ (1); $S_n = \frac{30}{2}(4 + 91) = 1425$ (2).	1+2

Mistakes to avoid:

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- Writing $S_n = \frac{n}{2}[a + (n-1)d]$; the bracket needs $2a$.
- Taking d as $a_1 - a_2$; it is $a_2 - a_1$ (a decreasing AP has negative d).
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